



Project Documentation

ChatCodeX

Group: Code Crusaders



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Introduction:

Project Title: ChatCodeX

Team Name: Code Crusaders

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Institution/Organization: Sukkur IBA University

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Abstract

This project aims to develop a Java-based messaging application with a graphical user interface (GUI) using Swing and AWT libraries. The application focuses on efficiently managing contacts and chat messages using data structures like LinkedList, Array, and Queue. The project will be implemented by a group of students as part of the Data Structures course under the guidance of Sir Abbas Mehdi.

Key Objectives:

- Develop a messaging application with two mobile screens (Person A and Person B).
- Implement functionalities to message each other, save new contacts, and search and delete contacts with chat.
- Ensure efficient data structures are used for managing contacts and messages.

Summary of Outcomes:

The application successfully demonstrates the use of various data structures for managing contacts and messages. It includes features like sending and receiving

messages, undoing the last message, searching messages, and toggling between themes.

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Introduction

Background and Context: The project is developed as part of the Data Structures course at Sukkur IBA University. The objective is to create a practical application demonstrating the use of different data structures.

Problem Statement: Develop a user-friendly messaging application that efficiently manages contacts and chat messages.

Objectives and Scope:

Develop a GUI-based messaging application.

Implement features like sending messages, saving contacts, searching and deleting contacts.

Use LinkedList, Array, and Queue for managing contacts and messages.

Importance of the Project: This project provides hands-on experience in developing a real-world application using Java and implementing data structures, which is crucial for understanding their practical applications.

System Requirements

Functional Requirements:

Two mobile screens for messaging between Person A and Person B.

Ability to send, receive, and display messages.

Functionality to add, search, and delete contacts.

Undo the last sent message.

Toggle between dark and light themes.

Non-Functional Requirements:

User-friendly interface.

Efficient data management.

Hardware and Software Requirements:

Hardware: Standard computer/laptop with necessary specifications.

Software: Eclipse IDE, Visual Studio Code, Java Development Kit (JDK), Microsoft Copilot.

Implementation

Description of Modules or Components:

ChatBoxGUI: Manages the chat interface and functionalities like sending messages, undoing last message, searching messages, displaying chat history, and toggling themes.

New: Manages the home page and adding new contacts.

Code Structure and Explanation (Optional):

Code is divided into two main classes: ChatBoxGUI and New.

Integration Process:

The two classes work together to provide a seamless user experience for messaging and contact management.

Results and Discussion

Key Outcomes of the Project:

Successfully implemented a messaging application with all required functionalities.

Demonstrated the use of LinkedList, Array, and Queue for managing data.

Comparison with Objectives:

The project met all key objectives, including efficient data management and a user-friendly interface.

Challenges Faced and How They Were Overcome:

Challenge: Implementing the undo functionality.

Solution: Utilized a Stack to keep track of messages for the undo feature.

References

Swing and AWT libraries for Java

Java.util package documentation for LinkedList and Queue

Eclipse IDE and Visual Studio Code documentation